



HELLENIC REPUBLIC

**National and Kapodistrian  
University of Athens**

— EST. 1837 —

**School of Medicine**

**MSc Applied Biomechanics in Orthopaedics**

**2019 - 2020**

**Postgraduate Thesis:**

**THE ROLE OF INERTIAL SENSORS IN THE  
FUNCTIONAL ASSESSMENT OF THE KNEE AFTER  
ANTERIOR CRUCIATE LIGAMENT  
RECONSTRUCTION SURGERY.  
A SYSTEMATIC REVIEW.**

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**Athens, 2022**

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The present dissertation was prepared in the context of the studies for the acquisition of a Postgraduate Specialization Diploma in "Applied Biomechanics in Orthopaedics" awarded by the School of Medicine of the National and Kapodistrian University of Athens.

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## **Acknowledgements**

I would like to express my gratitude to my primary supervisor, Prof Elias Tsepis, who guided me throughout this project. I would also like to thank the teaching staff of the postgraduate program for the valuable knowledge they imparted to me, as well as the secretariat for the technical support. Finally, I would like to thank my wife Despina for her unconditional support in these two very intense academic years.

# ABSTRACT

**Objective:** This article's primary purpose is to systematically review the literature on whether portable and affordable inertial sensors can reveal reliable inter-limb kinematic asymmetries in ACL reconstructed (ACLR) patients.

**Design:** Systematic review following PRISMA-DTA guidelines

**Methods:** We conducted a database search of PUBMED, CINAHL, SCOPUS, and SPORTDISCUS, checking for publications that compared the use of inertial sensors to motion analysis systems, assessing subjects with ACLR during a dynamic test or walking. The reviewers appraised every eligible study for its methodological quality according to the QUADAS-2 checklist.

**Results:** Initially, 279 articles were retrieved and finally, five met the inclusion criteria. In four studies, the extracted features from inertial sensors revealed significant inter-limb differences in ACLR patients. Some of these features were strongly correlated with metrics acquired from the motion analysis systems.

**Conclusion:** Inertial sensors can provide comparable information to motion analysis systems during dynamic knee assessment after ACLR surgery. This has significant potential for clinical use, enhancing patients' functional assessment and the monitoring of the rehabilitation process.

**Key words:** inertial sensors, knee assessment, ACL, diagnostic accuracy

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## **ABBREVIATIONS**

ACL: Anterior Cruciate Ligament

ACLR: Anterior Cruciate Ligament Reconstruction

IMU: Inertial Measurement Unit

KAV: Knee Angular Velocity

KMom: Knee Extensor Moment

KPow: Knee Power Absorption

MEMS: Micro-Electro-Mechanical Systems

MIMU: Magnetic Inertial Measurement Unit

PRISMA-DTA: Preferred Reporting Items for Systematic Reviews and Meta-Analyses  
for Diagnostic Test Accuracy

QUADAS-2: Quality Assessment of Diagnostic Accuracy Studies

SAV: Shank Angular Velocity

SLL: Single Limb Loading

TAV: Thigh Angular Velocity

# **1. INTRODUCTION**

Human motion analysis has a significant value in medicine, especially in orthopedics and sports medicine. The measurement of movement and how the forces are applied during a functional task can be valuable for appraising various surgical or therapeutic treatments (An, 1984).

The study of human movement has become an area of interest since the earliest times. During the renaissance, the concurrent evolution of science and mathematics settled the foundations for experimental observation. The unprecedented evolution of computer technology has provided us with sophisticated systems to capture human motion (Colyer et al., 2018). The gold standard for this purpose is a system consisting of cameras and retroreflective markers attached to the human body's anatomical landmarks, providing 3D visualization (Baker et al., 2007). Moreover, force measurement platforms in the ground provide the reaction forces during motion, and through inverse dynamics, can calculate net joint torques (Aurbach et al., 2017). Lastly, EMG recordings produce accurate signals of electrical muscle activity (Raez et al., 2006).

## **1.1 Optical motion capture systems**

The term motion capture includes many systems for quantitatively recording body movements. They can be divided into two major categories, optical and non-optical (Roggio et al., 2021). The first category is camera-based systems. These derive motion data from video recordings with special cameras. In the marker-based motion detection systems, multiple markers, active (luminous) or passive (reflective), are placed on the subject's prespecified anatomical landmarks according to various anthropometric models and a digital 3D representation is acquired. Then the video is captured from multiple angles using multiple cameras. The markers' positions define the body's segments coordinates in space and through mathematical algorithms and geometrical equations, the kinematics variables for the joints of interest are calculated in all movement planes (frontal, sagittal, and transversal). This is the most flexible and standard method used in the industry (Oppelt et al., 2019).

A markerless motion detection system has also been developed. There is no need to mark the body here. The software can track the subject's characteristics and infer relative motion from the video. Those movement data can be mapped to existing frameworks such as 3D models of the human body, hands, and face. Markerless systems can be based on regular video cameras, but it is also common to use depth-sensitive cameras. Such cameras use patterns of near-infrared light to recognize depth.

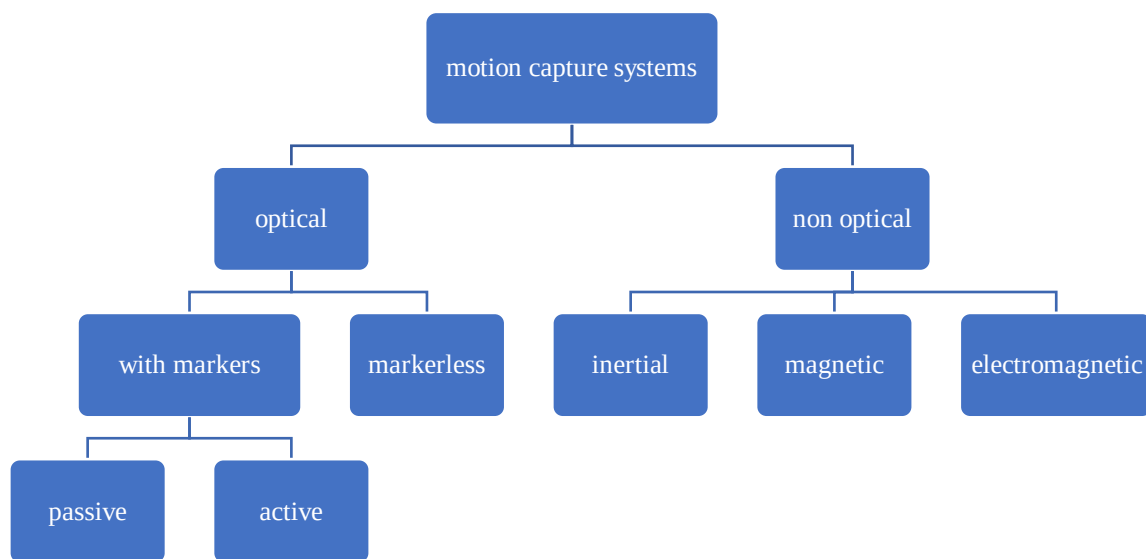
However, although these are still the gold-standard systems, they also have considerable disadvantages; establishing a gait laboratory requires equipment purchases that average \$300,000 maintenance contracts for hardware and software (\$30–\$50,000) and full-time laboratory personnel. They are also time-consuming and confined to a small collection space (Simon et al., 2004).

## **1.2 Non optical motion capture systems**

The second category, sensor-based systems, gets more detailed motion data from a small number of specific points (Andriacchi et al., 2006). Lately, new technology acquired from aerospace, industrial and robotic engineering appears a promising development in the field of human motion analysis. Small, low-powered electromechanical sensors may bridge the gap between large laboratory systems and clinical systems (Cuesta-Vargas et al., 2010).

These Micro-Electro-Mechanical Systems (MEMS) use accelerometers, magnetometers, and gyroscopes. Linear acceleration sensors are exclusively able to measure the acceleration along a specific direction. These transducers can be classified into one- two- and three-axis accelerometers depending on the number of sensitivity directions. When information regarding the device's rotation in space is required, as in gait analysis, angular velocity sensors (gyroscopes) must be added. A magnetometer measuring the earth's magnetic field orientation is often added to the sensor assembly to allow for easier re-calibration of the device and compensate for gyroscope drift issues. A device of a three-axis gyroscope and a three-axis accelerometer is called an Inertial Measurement Unit (IMU). When a magnetometer is also used, the device is known as MIMU (Magnetic Inertial Measurement Unit) (Petraglia et al., 2019).

Worn on the body segments, they provide motion information measured directly on the body, and they can be used for in-field measurements. Sensors are usually placed on each recorded body segment to derive joint kinematics. Rather than requiring accurate positioning, the user calibrates the IMUs according to the segment's coordinate system. Therefore, the placement procedure is less time-consuming compared to an optical system and their use requires much less expertise (Bleser et al., 2017).



**Figure 1.1: Types of motion capture systems**

Numerous studies have incorporated IMUs for assessing gait, functional movements, and sports activities throughout the literature. Gait analysis is the most commonly evaluated activity (Weygers et al., 2020). Compared with an optical motion capture system, wearable sensors can detect clinically relevant parameters with excellent test-retest reliability for all spatio-temporal metrics (Teufl, 2019; Washabough, 2017; Cho, 2018). 3D joint kinematics during dynamic tasks can also be assessed with a range of motion error below 3° (Teufl et al., 2019).

There is also an electromagnetic tracking system consisting of a transmitter and a sensor. The transmitter emits a low-frequency magnetic field, and a sensor placed on the moving

object measures the strength of each of these magnetic fields. From the field strength data for each axis, you can calculate the position and orientation of the sensor (Stelzer et al, 2004).

### **1.3 Anterior Cruciate Ligament**

Muscles, ligaments, and menisci supply the knee joint with rigidity and stability, allowing it to transfer enormous loads while performing numerous everyday duties (Markolf et al., 1976). The anterior cruciate ligament is the primary static stabilizer against the anterior translation of the tibia on the femur (Woo et al., 2006). It plays a vital role as a guide of the screw-home mechanism of the knee. The tibia internally rotates when the knee flexes and externally rotates as the femoral condyles roll and glide on the tibial plateau as the knee extends—the knee joint locks in a position of maximum stability during full extension (close-packed position) (Moglo et al., 2005).

For active individuals with an ACL rupture, ligament repair is the current standard of therapy (Marx et al., 2003). When designing a personalized post-surgical rehabilitation protocol for patients with ACLR, numerous factors should be considered, including the required return activity level, time to surgical intervention, type of surgery and graft choice, post-surgical muscle strength, and proprioceptive deficits (Della Villa et al., 2015). Despite the high functional outcomes after reconstructive surgery, biomechanical abnormalities and movement asymmetries can linger (Dunn et al., 2010). Biomechanical and neuromuscular control deficits and imbalances are closely connected to secondary injury (Paterno et al., 2010).

### **1.4 Biomechanical Asymmetries after ACLR**

Several biomechanical studies examined the kinematic and kinetic characteristics of ACLR patients. The ACLR groups had lower peak knee flexion angles, external knee flexion moments, and external knee adduction moments during walking (Kaur et al., 2016). This movement pattern during the loading phase of walking has been associated with cartilage deterioration in the knee and may have a role in the progression of osteoarthritis following ACLR (Slater et al., 2017).

Tibial rotation was observed to be substantially bigger in the ACLR knee compared to the contralateral intact leg and the healthy control group in a subsequent examination with a higher demand activity (stair descent and pivoting) (Ristanis et al., 2003). When combining

evidence, the most altered variables after ACLR are knee flexion motion and internal knee extension moment, which are consistently smaller in the reconstructed limb while running (Pairot-de-Fontenay et al., 2019).

Asymmetry was found more frequently in kinetic variables than in kinematic variables. The asymmetry between limbs for peak vertical ground reaction force, peak knee-extension moment, and loading rate during double-limb landings, as well as the mean knee-extension moment and knee energy absorption during both double- and single-limb landings, appeared to be significantly affected by anterior cruciate ligament reconstruction (Hughes et al., 2020).

### **1.5 Quadriceps Strength Deficits**

Following ACL rupture and surgery, significant quadriceps atrophy progresses rapidly (Williams et al., 2005). Strength impairments ranging from 2% to 20% have been documented in participants more than two years after ACL surgery (Palmieri-Smith et al., 2008). The underlying causes for such weakness are still unknown. Researchers have also proposed that the inability to engage the quadriceps muscle fully (i.e., voluntary activation failure) is a crucial source of persistent quadriceps weakness in persons who have had ACL repair (Hart et al., 2010). This concept, however, has not been validated. Graft-related muscular impairments occur following ACL repair. A patellar tendon transplant is related to extensor deficits, whereas hamstring grafts are associated with flexion abnormalities (Petersen et al., 2014). One of the most crucial rehabilitation goals following ACLR is to restore lower extremity muscular strength (Buckthorpe et al., 2019).

Regarding the lower limb, knee osteoarthritis and total knee arthroplasty have been the two most common musculoskeletal issues where inertial sensors are used as a diagnostic tool for measuring risk factors and recovery.

### **1.6 Objectives**

This article's primary purpose is to systematically review the literature on the inertial sensors' use for functional assessment of patients who have undergone ACL reconstructive surgery. To investigate whether IMUs can distinguish asymmetries in the function of the reconstructed knee and whether metrics derived from IMUs can significantly correlate with metrics from the gold standard motion analysis systems.

## 2. METHODS

This systematic review is based on Preferred Reporting Items for Systematic Review and Meta-Analyses (PRISMA) Extension for Diagnostic Test Accuracy for reporting.

### 2.1 Information sources and search strategy

We conducted systematic research in the following electronic databases: PubMed, Scopus, Cinahl, Sportdiscus. The Boolean strategy is depicted in the table (2.1). Moreover, we examined reference lists to retrieve relevant studies.

| <b>Table 2.1.</b> Boolean search strategy |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Database</b>                           | <b>Keywords PubMed, Scopus, Cinahl, Sportdiscus</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                           | ((inertial sensors) OR (IMUs)) AND ((anterior cruciate ligament) OR (ACL)) AND (deficits)<br>((inertial sensors) OR (IMUs)) AND ((anterior cruciate ligament) OR (ACL)) AND (assessment)<br>((inertial sensors) OR (IMUs)) AND ((anterior cruciate ligament) OR (ACL)) AND (injury)<br>((inertial sensors) OR (IMUs)) AND ((anterior cruciate ligament) OR (ACL)) AND (evaluation)<br>((inertial sensors) OR (IMUs)) AND ((anterior cruciate ligament) OR (ACL)) AND (asymmetries)<br>(accelerometry OR accelerometer OR gyroscope OR inertial sensor OR inertial measurement unit OR wearable sensor OR wearable system OR wearable device OR IMU OR MEMS) AND (anterior cruciate ligament OR ACL) |

### 2.2 Study selection

The articles were initially imported into Endnote, and after removing duplicates, they were screened by title and abstract. Two authors independently extracted data from the studies according to the inclusion criteria.

### 2.3 Eligibility criteria

For the analysis, we included any study or trial that uses inertial sensor units to evaluate the knee function during physical activities or special tests regarding the lower limb as long as they are referred to a population undergoing anterior cruciate ligament reconstructive surgery (ACLR). This analysis could be conducted in a clinical setting or a human movement analysis lab but with the concurrent use of a gold standard motion analysis system. A summary of inclusion and exclusion criteria is given in table 2.2

| <b>Table 2.2.</b> Inclusion, exclusion criteria                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                 |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Inclusion criteria</b>                                                                                                                                                                                                                                          | <b>Exclusion criteria</b>                                                                                                                                                                                                                                                                                                                                                       |
| <ul style="list-style-type: none"> <li>-Studies including the use of inertial sensors and motion analysis systems for evaluating knee function after ACLR surgery</li> <li>-Studies published in English</li> <li>-Studies published in 2000 or earlier</li> </ul> | <ul style="list-style-type: none"> <li>-Conference abstracts</li> <li>-Systematics reviews</li> <li>-Studies that do not refer to ACL injuries</li> <li>-Studies that use inertial sensors for passive evaluation of the knee joint and not through dynamic testing</li> <li>-Studies using cadaveric, robotic, or animal models</li> <li>-Published study protocols</li> </ul> |

## 2.4 Assessment of risk of bias

The retrieved documents were assessed for methodological quality using the QUADAS-2 checklist in four domains: patient selection, index test, reference standard, flow and timing. We evaluated all four domains for risk of bias, and the first three also regarding applicability.

## 2.5 Diagnostic accuracy measures and synthesis of results

The data extraction from the selected studies was focused on the patients' characteristics, the type, the number and the anatomical positioning of sensors used. The accuracy and reliability measures of the IMUs compared to the gold standard optoelectronic 3D motion analysis system and the correlation of the metrics between the two systems were also primary areas of interest.



### 3. RESULTS

The initial search retrieved 279 publications. After removing duplicates, remained 157 studies. Through title and abstract reading, we included 35 articles for full-text reading. Finally, five studies were included in this review.

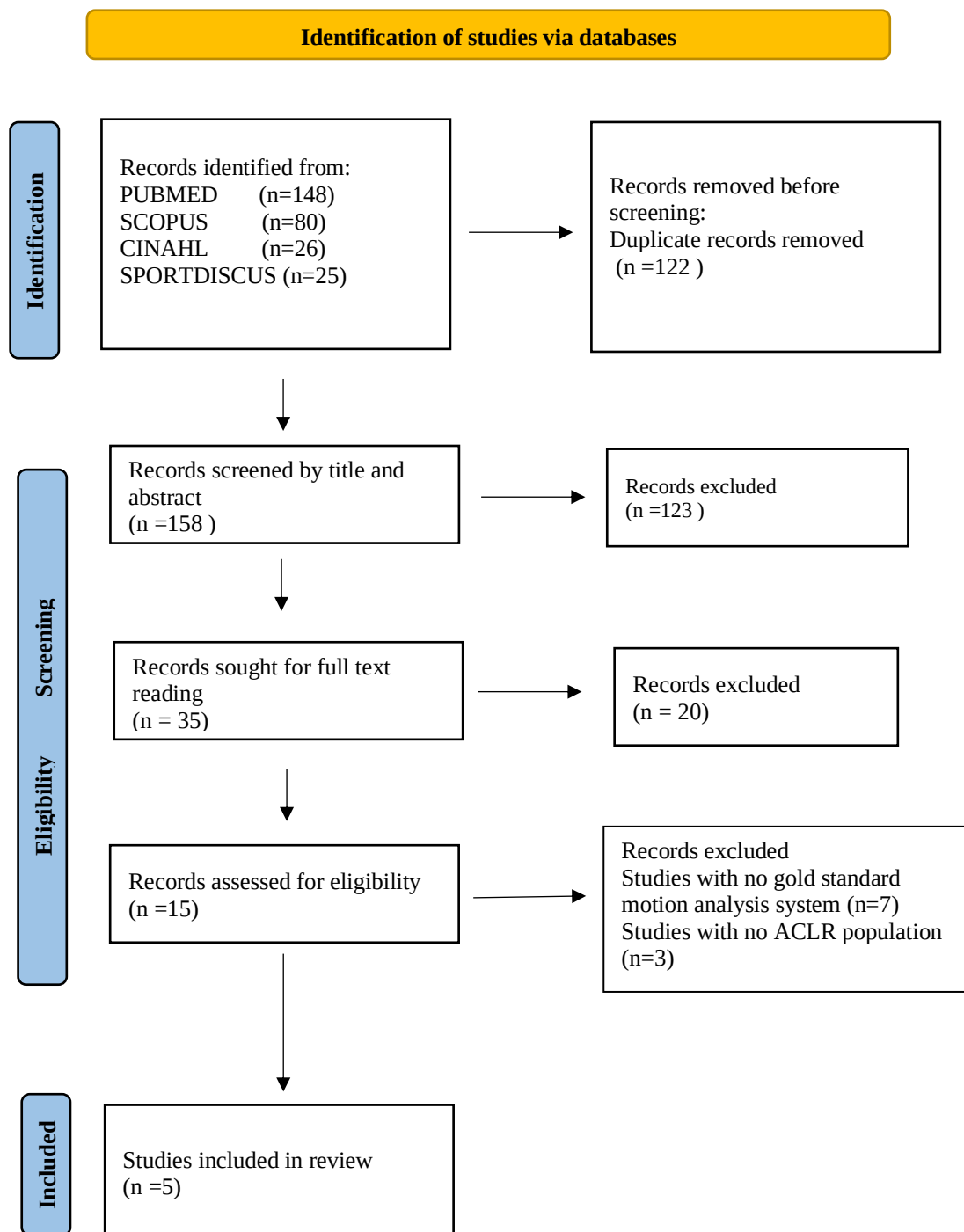


Figure 3.1. PRISMA literature review flow diagram

### 3.1 Demographic data

A total of 109 subjects between 17 and 40 years old were examined. Only one study included a healthy control group consisting of 17 people. Ninety-two had undergone ACL reconstruction, with time since surgery ranging from three months to three and a half years. The majority of the patients had undergone surgery four months ago (Table 3.1).

| <b>Table 3.1.</b> Population summary for the studies |                  |        |      |            |                    |
|------------------------------------------------------|------------------|--------|------|------------|--------------------|
| Study                                                | Age              | ACLR   |      | Control    | Time Since Surgery |
|                                                      |                  | Female | Male |            |                    |
| <b>Patterson et al. (2014)</b>                       | 23,7 ± 3,12 yrs. | 17     | -    | 17 females | 3,50 ± 3,25 yrs.   |
| <b>Sigward et al. (2016)</b>                         | 26,8 ± 12,4 yrs. | 14     | 5    | -          | 96,7 ± 16,8 days   |
| <b>Pratt et al. (2018)</b>                           | 28,8 ± 11,2 yrs. | 12     | 9    | -          | 5,1 ± 1,5 months   |
| <b>Pratt et al. (2018)</b>                           | 28,8 ± 11,2 yrs. | 12     | 9    | -          | 5,1 ± 1,5 months   |
| <b>Havens et al. (2018)</b>                          | 29 ± 12 yrs.     | 7      | 7    | -          | 20,3 ± 7,1 weeks   |

### 3.2 Tested devices - index

Three-axis accelerometers, gyroscopes, and magnetometers contained in a portable device were tested in all eligible studies. Most authors utilized the signal from gyroscopes to capture the angular velocity, and only one used the accelerometer's output as the primary variable of interest. The number of IMUs used also varied amongst the studies, with three using two IMUs, and four using four IMUs. Precise positioning and additional technical information are provided in table 3.2.

| <b>Table 3.2.</b> Inertial Sensors summary used in studies |                                          |             |                |                 |               |
|------------------------------------------------------------|------------------------------------------|-------------|----------------|-----------------|---------------|
| Study                                                      | Model                                    | Sensors     | Number of IMUs | Position        | Sampling Rate |
| <b>Patterson et al. (2014)</b>                             | Xsens MTx G25, Enchende, The Netherlands | A3 + G3     | 2              | shank bilateral | 100 Hz        |
| <b>Sigward et al. (2016)</b>                               | Opal APDM, Portland, OR, USA             | A3 + G3 +M3 | 2              | shank bilateral | 128 Hz        |

| <b>Table 3.2.</b> Inertial Sensors summary used in studies (Continued)        |                              |                |                        |                           |                      |
|-------------------------------------------------------------------------------|------------------------------|----------------|------------------------|---------------------------|----------------------|
| <b>Study</b>                                                                  | <b>Model</b>                 | <b>Sensors</b> | <b>Numbers of IMUs</b> | <b>Position</b>           | <b>Sampling Rate</b> |
| <b>Pratt et al. (2018)</b>                                                    | Opal APDM, Portland, OR, USA | A3 +G3 + M3    | 2                      | thigh bilateral           | 128 Hz               |
| <b>Pratt et al. (2018)</b>                                                    | Opal APDM, Portland, OR, USA | A3 + G3 +M3    | 4                      | thigh and shank bilateral | 128 Hz               |
| <b>Havens et al. (2018)</b>                                                   | Opal APDM, Portland, OR, USA | A3 + G3 +M3    | 4                      | thigh and shank bilateral | 128 Hz               |
| A3: triaxial accelerometer; G3: triaxial gyroscope; M3: triaxial magnetometer |                              |                |                        |                           |                      |

### 3.3 Reference standard

All eligible studies used a gold standard motion analysis system to obtain and compare results with IMUs. Patterson et al. (2014) used a three-camera system and the rest researchers a more advanced setup with 14 cameras. Furthermore, motion capture systems were coupled with force platforms in all studies. (Table 3.3)

| <b>Table 3.3.</b> Motion analysis systems summary used in studies |                               |                   |                          |
|-------------------------------------------------------------------|-------------------------------|-------------------|--------------------------|
| <b>Study</b>                                                      | <b>Motion Analysis System</b> | <b>Cameras</b>    | <b>Force Platforms</b>   |
| <b>Patterson et al. (2014)</b>                                    | CODA                          | 3 MPX30 at 200 Hz | 2 AMTI at 1000 Hz        |
| <b>Sigward et al. (2016)</b>                                      | Qualysis Inc. or              | 11 at 250 Hz      | AMTI at 1500 Hz or       |
|                                                                   | BTS Smart - DX                | 14 at 340 Hz      | BTS P-6000 DX at 1360 Hz |
| <b>Pratt et al. (2018)</b>                                        | Qualysis Inc.                 | 11 at 250 Hz      | AMTI at 1500 Hz or       |
|                                                                   | BTS Smart - DX                | 14 at 340 Hz      | BTS P-6000 DX at 1360 Hz |
| <b>Pratt et al. (2018)</b>                                        | Qualysis Inc.                 | 11 at 250 Hz      | AMTI at 1500 Hz or       |
|                                                                   | BTS Smart - DX                | 14 at 340 Hz      | BTS P-6000 DX at 1360 Hz |
| <b>Havens et al. (2018)</b>                                       | BTS Bioengineering Corp.      | 14 at 340 Hz      | 4 BTS at 1360 Hz         |

### 3.4 Test Tasks

The test procedures to investigate the load differences in ACLR subjects were different amongst the studies. Patterson et al. (2014) and Sigward et al. (2016) incorporated walking, Havens et al. (2018) running, and Pratt et al. (2018) a single limb loading task. (Table 3.4)

| <b>Table 3.4.</b> Test tasks summary |                                                                                                                                                                                                                                                                                                                                                                                                            |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>STUDY</b>                         | <b>TEST TASKS</b>                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Patterson et al. (2014)</b>       | Walk barefoot across 15m walkway at their self-selected average walking speed.                                                                                                                                                                                                                                                                                                                             |
| <b>Sigward et al. (2016)</b>         | Walk at their comfortable pace across a 10m path.                                                                                                                                                                                                                                                                                                                                                          |
| <b>Pratt et al. (2018)</b>           | Dynamic Single limb loading task.<br>Participants were standing on both feet on a single force platform in front of a target where was another force platform. The distance they had to cover was normalized to each participant's tibia length. They leaped forward to the mark onto a single limb, lowering themselves as far as they could, and then returned to the starting position without pausing. |
| <b>Havens et al. (2018)</b>          | Participants were running at a self-selected speed for 15m.                                                                                                                                                                                                                                                                                                                                                |

### 3.5 Risk of bias and applicability concerns

Considering the risk of bias and applicability, we evaluated each article according to the QUADAS-2 tool. Patient selection in all studies was rated as high risk because there was not clear how the collection was performed. Patterson et al. (2014) incorporated a control group, which could have exaggerated the diagnostic accuracy of the index test.

In the index test and reference standard domains, we rated the risk of bias unclear in all the studies. The main concern was if the results of the one could have influenced the interpretation of the other. Flow and timing domain was rated as low risk of bias, but not all subjects received the same reference standard in three studies. Finally, all studies were rated as low in terms of the applicability domain.

### 3.6 IMUs Diagnostic accuracy

In all eligible studies, the use of IMUs revealed significant differences between limbs or between the ACLR group and control in some metrics. In two studies, inertial sensors were tested during walking. Patterson et al. (2014) found no differences in temporal parameters between ACLR and control. Still, they revealed significant differences in three gyroscope variables, peak shank angular velocity (SAV) at midstance, minimum rotation rate at initial contact, and post heel strike shank adjustments. All the gyroscope extracted features significantly differed between the ACLR and the control group and were positively correlated.

On the other hand, Sigward et al. (2016) found no differences between limbs during stance and swing phase but found differences at peak SAV during loading response. The review included two studies from Pratt et al. (2018). Both of them used inertial sensors during a dynamic single limb loading task. Metrics were taken during the deceleration phase. The first one used only two sensors at thighs and found asymmetries at peak thigh angular velocity (TAV). The second one placed two more sensors at the shanks and revealed differences between knee angular velocity (KAV), TAV and SAV for the ACLR limbs. Finally, Havens et al. (2018) used the metrics from the accelerometer but found no differences between limbs in the thigh and shank axial acceleration during running.

**Table 3.5.** Summary of IMUs' extracted variables and inter-limb differences

| STUDY                          | IMUS                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                         |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                | Extracted Variables                                                                                                                                                                                                                                                                                                             | Between Limb Differences                                                                                                                                                |
| <b>Patterson et al. (2014)</b> | Gait cycle duration<br>Stance time<br>Swing time<br>Double support time<br>Shank rotation rate (toe-off)<br>Shank rate of change (initial swing)<br>Peak shank rotation rate (swing)<br>Shank rotation rate (initial contact)<br>Post initial contact shank rotation rate variance<br>Shank rotation rate variance (mid-stance) | <b><i>Smaller on ACLR:</i></b><br>Peak shank rotation rate during swing<br>Shank rotation rate at initial contact<br>Post- initial contact shank rotation rate variance |
| <b>Sigward et al. (2016)</b>   | Peak SAV at mid-swing<br>Peak SAV at heel strike<br>Peak SAV at loading response<br>Between limb ratio of peak SAV                                                                                                                                                                                                              | <b><i>Smaller on ACLR:</i></b><br>Peak SAV during loading response                                                                                                      |
| <b>Pratt et al. (2018)</b>     | Peak TAV<br>Peak TAV ratios                                                                                                                                                                                                                                                                                                     | <b><i>Smaller on ACLR:</i></b><br>TAV ratios                                                                                                                            |
| <b>Pratt et al. (2018)</b>     | Peak KAV<br>Peak TAV<br>Peak SAV                                                                                                                                                                                                                                                                                                | <b><i>Smaller on ACLR:</i></b><br>KAV<br>TAV<br>SAV                                                                                                                     |
| <b>Havens et al. (2018)</b>    | Shank axial acceleration<br>Thigh axial acceleration                                                                                                                                                                                                                                                                            | No differences                                                                                                                                                          |

### 3.7 Motion analysis systems' diagnostic accuracy

Optoelectronic motion analysis systems are the gold standard for revealing asymmetries in all five studies. Patterson et al. (2014) found that the ACLR group displayed lower peak knee adduction moments during early stance. Also, the ACLR group showed small knee joint angular kinematic differences during the swing phase (more extended and adducted knee position). No temporal or spatial parameter alterations were noted between groups. Sigward et al. (2016) found differences in peak knee internal extensor moment during loading response (approximately 20% of stance) but no differences at temporal events of gait. In their first study, Pratt et al. (2018) calculated net joint power, peak knee power absorption and knee power symmetry ratios and found the last smaller on the ACLR limb. In the second study, Pratt et al. (2018) recorded peak knee extensor moment, knee angular velocity, thigh angular velocity and shank angular velocity. All features were smaller on the ACLR limb. Finally, Havens et al. (2018) utilized peak vertical GRF, peak knee extensor moment, sagittal plane peak knee power absorption, peak knee joint angular velocity and knee range of motion. Compared to the non-surgical limb, individuals exhibited smaller knee joint biomechanical variables.

| <b>Table 3.6.</b> Summary of motion analysis systems' extracted variables inter-limb differences |                                                                                                                                                    |                                                                                           |
|--------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| <b>STUDY</b>                                                                                     | <b>MOTION ANALYSIS SYSTEM</b>                                                                                                                      |                                                                                           |
|                                                                                                  | <b>Extracted Variables</b>                                                                                                                         | <b>Between Limb Differences</b>                                                           |
| <b>Patterson et al. (2014)</b>                                                                   | Gait cycle duration<br>Stance time<br>Swing time<br>Double support time<br>Knee angular kinematic<br>Kinetic data                                  | <b>Smaller on ACLR:</b><br>Peak knee adduction moment<br>Normalized knee adduction moment |
| <b>Sigward et al. (2016)</b>                                                                     | Peak KMom during loading response<br>Between limb ratio of peak KMom                                                                               | <b>Smaller on ACLR:</b><br>Peak KMom                                                      |
| <b>Pratt et al. (2018)</b>                                                                       | Net joint power<br>Peak knee power absorption<br>Knee power symmetry ratios                                                                        | <b>Smaller on ACLR:</b><br>Knee power symmetry ratios                                     |
| <b>Pratt et al. (2018)</b>                                                                       | Peak knee power absorption<br>Peak knee extension moment<br>KAV, TAV, SAV                                                                          | All features smaller                                                                      |
| <b>Havens et al. (2018)</b>                                                                      | Peak vertical GRF<br>Peak knee Ext moment<br>Sagittal plane peak knee power absorption<br>Peak knee joint angular velocity<br>Knee range of motion | <b>Smaller on ACLR:</b><br>Sagittal plane knee range of motion<br>KMom<br>KPow<br>KAV     |

### 3.8 IMUs correlation with the Motion analysis system

All eligible studies integrated motion analysis systems into their experiments along with IMUs. In Patterson et al. (2014) study, the ACLR participants displayed significantly lower peak knee adduction moments at early stance. Still, the results did not confirm a correlation with the reduced shank rotation rate acquired from the IMUs during the swing phase. The second article testing walking by Sigward et al. (2016) found a positive correlation between knee extensor moment (KMom) and shank angular velocity during the loading response phase. Greater shank angular velocity was related to greater knee extensor moment. In the first article from Pratt et al. (2018), thigh angular velocity symmetry ratios (peak TAV in reconstructed limb divided by peak TAV in non-surgical limb) positively correlated with knee power ratio (knee power absorption in the reconstructed limb divided by knee power absorption in the non-surgical knee). Inertial sensor measurements showed no false positives (100% specificity) and a few false negatives (81,2% sensitivity). In the second article from Pratt et al. (2018), the statistical analysis indicated strong agreement between measurements systems for KAV, TAV, and SAV during single limb loading test (SLL). Furthermore, TAV was strongly correlated with knee power absorption (KPow) and moderately associated with knee extensor moment (KMom). Finally, Havens et al. (2018) found that limb differences in thigh axial acceleration were the only predictor of ground reaction force (GRF) differences and knee power absorption.

| <b>Table 3.7.</b> Summary of correlation between extracted variables of the two systems |                                    |                               |
|-----------------------------------------------------------------------------------------|------------------------------------|-------------------------------|
| <b>STUDY</b>                                                                            | <b>CORRELATION BETWEEN SYSTEMS</b> |                               |
|                                                                                         | <b>IMUs</b>                        | <b>Motion analysis system</b> |
| <b>Patterson et al. (2014)</b>                                                          | No correlation                     |                               |
| <b>Sigward et al. (2016)</b>                                                            | Between limb ratio of SAV          | Between limb ratio of KMom    |
| <b>Pratt et al. (2018)</b>                                                              | TAV symmetry                       | KPow symmetry                 |
| <b>Pratt et al. (2018)</b>                                                              | TAV                                | KPow and KMom                 |
| <b>Havens et al. (2018)</b>                                                             | TAA                                | GRF and KPow                  |

## 4. DISCUSSION

The purpose of this review is to investigate whether the use of IMUs can reliably reveal loading differences after an ACLR surgery. We included five studies, and a total of 109 participants were analyzed. Angular velocities derived from IMUs were the primary outcome measure. Two studies (Patterson, 2014; Sigward, 2016) tested these portable devices during walking and found lower SAV in the ACLR limb than in the healthy limb or control group. In particular, ACLR limbs depicted differences in SAV during the swing, initial contact, and loading response. Pratt et al. (2018) presented two articles with the same SLL task measuring angular velocities during the deceleration phase. In both of them, SAV and TAV were lower. In one study (Havens et al., 2018), thigh and shank axial acceleration was the prime variable of interest, but it did not differ during running.

Regarding the association between the metrics from IMUs and the gold standard motion analysis system, thigh and shank angular velocities were positively correlated with knee extensor moment and knee power absorption. Pratt et al. (2018) also demonstrated that thigh angular velocity was more predictive of KPow and KMom than SAV. Lastly, Havens et al. (2018) correlated thigh axial acceleration with GRF and KPow.

To our knowledge, this is the first review focusing on the use of IMUs in assessing knee function after ACLR surgery compared with gold standard motion analysis systems. Petraglia et al. (2019) recently reviewed the application of IMUs versus traditional methods in gait analysis, but the included studies did not have any participants with orthopedic knee conditions.

Another scoping review was issued in 2019 by Small et al. investigating wearable sensors' methodology and clinical application to evaluate patient recovery and functional outcomes following knee arthroplasty. This work provides a comprehensive image of the aforementioned aspects without a detailed comparison with the motion analysis systems.



## **4.1 Clinical implications and considerations**

Throughout the literature, researchers use IMUs to identify movement disorders across a broad spectrum of neurological and orthopaedical conditions. They have shown good agreement with classical gait analysis systems regarding the spatio-temporal parameters, even in real-life situations (Petraglia et al., 2019). Gait pattern studies have shown that the kinematics in the sagittal plane of the reconstructed limb return to the normal level of healthy subjects after six months (Gao, 2010; Leporace, 2012). Spatio-temporal variables are not suitable parameters to distinguish knee gait biomechanics of ACLR subjects from healthy ones and should not be used as benchmarks in determining return to sports after ACL reconstruction (Leporace et al., 2016).

Throughout the rehabilitation process, ACLR results in subtle and persistent kinematic and kinetic alterations in the knee and consequently in lower extremity movement patterns. These changes may not be detrimental for the ACL craft while walking, but they have been associated with the early onset of knee osteoarthritis (Gokeler, 2013; Slater, 2017).

In this review, two studies have confirmed that direct outputs from IMUs can reveal differences in shank angular velocities during walking even in individuals three years post-ACLR. Furthermore, Sigward et al. (2016) established a link between the kinematic differences detected with the IMUs and loading impairments assessed with the motion analysis system. Specifically, the lower shank angular velocity of the ACLR limb is correlated with the lower knee extensor moment. During the second rocker of gait, the tibia is rolled forward over the talus, rotating around the ankle joint (Mayich et al., 2014). At that point, the shank rotates faster than the femur, resulting in knee flexion, which coincides with a peak knee extensor moment generated by the activation of the quadriceps. The recovery of quadriceps strength is often incomplete even years after surgery (Krishnan et al., 2011). Detecting between limp differences in shank angular velocity during loading response with the help of IMUs may be helpful for the clinician to assess between limp deficits and address them in their everyday practice.

The most significant hurdle for sports clinicians is how to return the athlete to their original sport or any individual who has undergone an ACLR surgery at an even greater level of functional ability and, at the same time, significantly reducing the possibility of reinjury (Kaplan et al., 2019). Eighty-one percent of individuals return to some form of sport after surgery, 65% return to their previous level of sport, and 55% return to competitive sport (Ardern, 2014; Lai, 2018). Nevertheless, even after rehabilitation, between 35% and 45% of injured athletes will not return to perform at premorbid levels (Sepulveda et al., 2017). Moreover, there is an increased reinjury rate among young athletes who have undergone an ACLR; 1 in 4 sustain another ACL injury at some point in their career relatively early upon their return to high-risk sport (Wiggins, 2014; Paterno, 2014). Presently, there is still no gold standard to clear athletes for return to sport (RTS) after ACLR surgery. The clinical decision is multifactorial and based on a combination of physical and psychological factors (Cheney et al., 2020).

The evidence from the two studies from Pratt et al. (2018) suggests that IMUs can serve as a valuable tool to detect dynamic loading deficits of the knee when the patient is ready for a more challenging task like running. A previous study from Dowling et al. (2012) has reported a relationship between coronal plane thigh and shank angular velocities and knee adductor moments in a healthy population during a jump landing task. Pratt et al. (2018) correlated segment angular velocity to joint power (product of joint moment and joint angular velocity), exploring the power absorption during sagittal plane loading. These results imply that in the lack of force platforms and a marker-based motion capture system, inertial sensors and, more specifically, a sensor based on the thigh may be helpful in the clinic to quantify modified knee loading during a single limb loading task after an ACLR surgery.

At this point, two more studies can be mentioned even though they were not met the inclusion criteria. The first is from Setuain et al. (2015), which quantifies jumping biomechanics using direct mechanics-based approaches to simplify the measurement methodology. They studied the behavior of the center of mass during various jumping activities, taking into consideration the complete body as a single system of mass and inertia. For that reason, a single IMU was used in the L3 area where the human's center of gravity is thought to be. After identifying the various events of the jumping task (based on vertical velocity records), the linear acceleration, direction, and jumping

phase timings were assessed to determine the important parameters for each phase. Compared to their non-ACL injured counterparts, ACL-reconstructed athletes had higher supporting three-axis peak accelerations and worse jumping performance (expressed as the duration in seconds of the flight time phase of the jump).

The second one is from Dan et al. (2019). They predicted changes in limb kinetics during functional exercise that the current return to sports assessments does not identify. The current study aimed to investigate the effectiveness of this accelerometer, gyroscope system, and pressure sensor mat in detecting patients' kinetic differences prior to returning to sport after ACL surgery. The main factors of concern were varus and valgus alignment. ACL patients showed more significant deviations in the reconstructed leg when undertaking single-leg exercises such as the single-leg box drop and single-leg squat. There were no variations in alignment in the double leg activities; however, the force plate revealed that the reconstructed leg involves less force than the contralateral leg. The force plate also demonstrated a reduced flying time in the repaired leg during a single-leg hop test. IMUs did not find any asymmetrical variations in coronal alignment in double-leg exercise. This might be explained by the patient adjusting to accept less force through the repaired limb, but the force plate detects less force via the repaired leg in a double leg landing.

Inertial-based motion capture handles many limitations of traditional motion capture systems, such as marker occlusion, expensive equipment costs, and performing movements in a confined laboratory environment. The accuracy of IMU systems is mainly determined by the data fusion algorithms and pose estimation methodologies used to interpret human motion from raw IMU data. Also, there are several sources of possible error. To enhance IMU-derived spatiotemporal and kinematic estimations, the clinician must consider the effect of soft tissue artifacts, device placement, biomechanical modeling methods, and ferromagnetic interference within the environment. (Hindle et al., 2021).

## **4.2 Limitations**

The current study has a number of limitations. First of all, although we retrieved enough articles referring to the dynamic assessment of the knee after an ACLR surgery with the use of IMUs, only five met the eligible criteria. Most of the studies excluded from

this review lacked a concurrent evaluation by a gold standard optoelectronic system or did not incorporate an ACLR population. Secondly, the included studies were not uniform regarding the executed tasks and the outcome measures. Precisely, we encountered three different tasks that the authors used to compare knee function and various outcome measures. Additionally, it is unknown if the reported diagnostic accuracy applies to other phases of rehabilitation than those mentioned or is maintained with multiple dynamic tests. Future studies are needed to establish a more standardized test procedure and incorporate IMUs as a diagnostic tool.

### **4.3 Conclusion**

Inertial sensor measurement units can serve as a valuable instrument in the clinical quantification of dynamic knee loading asymmetries. In the absence of high-end motion analysis systems and force platforms, inertial sensors may be a practical alternative for analyzing knee function. Further development is needed to improve their techniques and their application in an ACLR population

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## ΠΕΡΙΛΗΨΗ

**Στόχος:** Ο πρωταρχικός σκοπός αυτού του άρθρου είναι η συστηματική ανασκόπηση της βιβλιογραφίας σχετικά με το εάν οι φορητοί και οικονομικά προσιτοί αδρανειακοί αισθητήρες, μπορούν να αποκαλύψουν με αξιοπιστία κινηματικές ασυμμετρίες μεταξύ των άκρων σε ασθενείς με ανακατασκευή Πρόσθιου Χιαστού Συνδέσμου (ΠΧΣ).

**Σχεδιασμός:** Συστηματική ανασκόπηση σύμφωνα με τις οδηγίες PRISMA-DTA

**Μεθοδολογία:** Πραγματοποιήσαμε συστηματική αναζήτηση στη βάση δεδομένων των PUBMED, CINAHL, SCOPUS και SPORTDISCUS, ελέγχοντας για δημοσιεύσεις που συνέκριναν τη χρήση αδρανειακών αισθητήρων με οπτοηλεκτρονικά συστήματα ανάλυσης κίνησης, αξιολογώντας άτομα με ανακατασκευασμένο (ΠΧΣ) κατά τη διάρκεια μιας δυναμικής δοκιμής ή περπατήματος. Κάθε επιλέξιμη μελέτη αξιολογήθηκε για τη μεθοδολογική της ποιότητα σύμφωνα με τη λίστα ελέγχου QUADAS-2

**Αποτελέσματα:** Αρχικά ανακτήθηκαν 279 άρθρα και τελικά πέντε πληρούσαν τα κριτήρια συμπερίληψης στην έρευνα. Τα εξαγόμενα χαρακτηριστικά από αδρανειακούς αισθητήρες αποκάλυψαν σημαντικές διαφορές μεταξύ των άκρων σε ασθενείς με ανακατασκευασμένο (ΠΧΣ) σε τέσσερις μελέτες. Ορισμένα από αυτά τα χαρακτηριστικά συσχετίστηκαν ισχυρά με μετρήσεις που αποκτήθηκαν από τα συστήματα ανάλυσης κίνησης.

**Συμπέρασμα:** Οι αδρανειακοί αισθητήρες μπορούν να παρέχουν συγκρίσιμες πληροφορίες με τα συστήματα ανάλυσης κίνησης κατά τη δυναμική αξιολόγηση του γόνατος μετά από χειρουργική επέμβαση (ΠΧΣ). Αυτό έχει σημαντικές δυνατότητες για κλινική χρήση, ενισχύοντας τη λειτουργική αξιολόγηση των ασθενών και την παρακολούθηση της διαδικασίας αποκατάστασης τους.

**Λέξεις κλειδιά:** αδρανειακοί αισθητήρες, αξιολόγηση γόνατος, πρόσθιος χιαστός, διαγνωστική ακρίβεια.